

Binary Semantic Segmentation

Land / Water

Computer Vision

Emanuele Romanzi

1935332



SAPIENZA
UNIVERSITÀ DI ROMA



Index

- › Context
- › Challenges
- › Approach
- › Proposed Solution
- › Results



Context

Semantic Segmentation

Semantic segmentation is a computer vision task that assigns a **semantic class to each pixel of an image**, producing a dense representation of the scene.

Unlike image classification, it preserves spatial information, identifying not only what is present in an image, but also where it is located.



(a) Source Image



(b) Semantic Segmentation



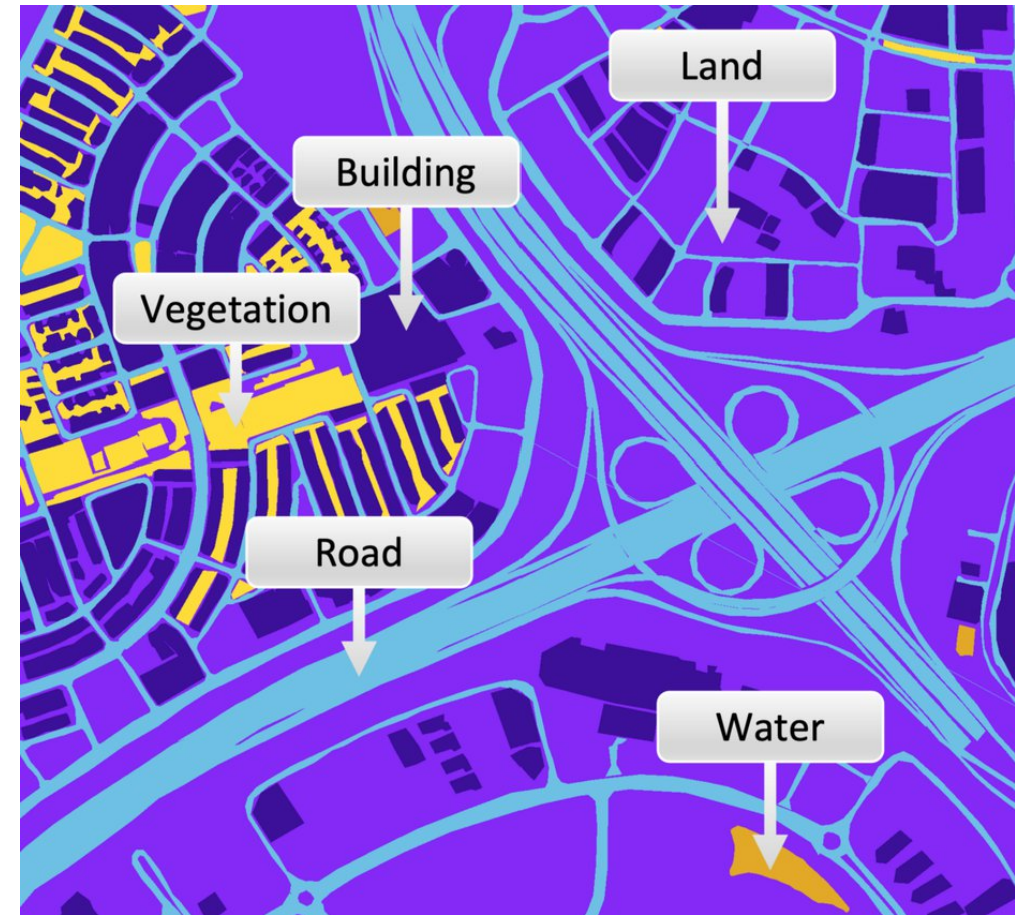
Context

Semantic Segmentation with Satellite Images

Satellite image segmentation **enables the pixel-level identification and mapping of different surface types**, such as water, vegetation, urban areas, etc.

Unlike conventional RGB images, **satellite imagery can provide information across multiple spectral bands**, capturing different portions of the electromagnetic spectrum.

This additional **spectral information can improve the discrimination between surface classes**.





Index

- › Context
- › Challenges
- › Approach
- › Proposed Solution
- › Results



Challenges

Spatial Data Leakage

Spatial data leakage occurs when geographically close samples are assigned to different training, validation or test sets.

Nearby satellite patches often share **similar spectral patterns and spatial features** making them highly spatially correlated.

As a result, **the model may indirectly benefit from information already observed during training**, leading to overly optimistic metrics that do not reflect its true generalization ability.





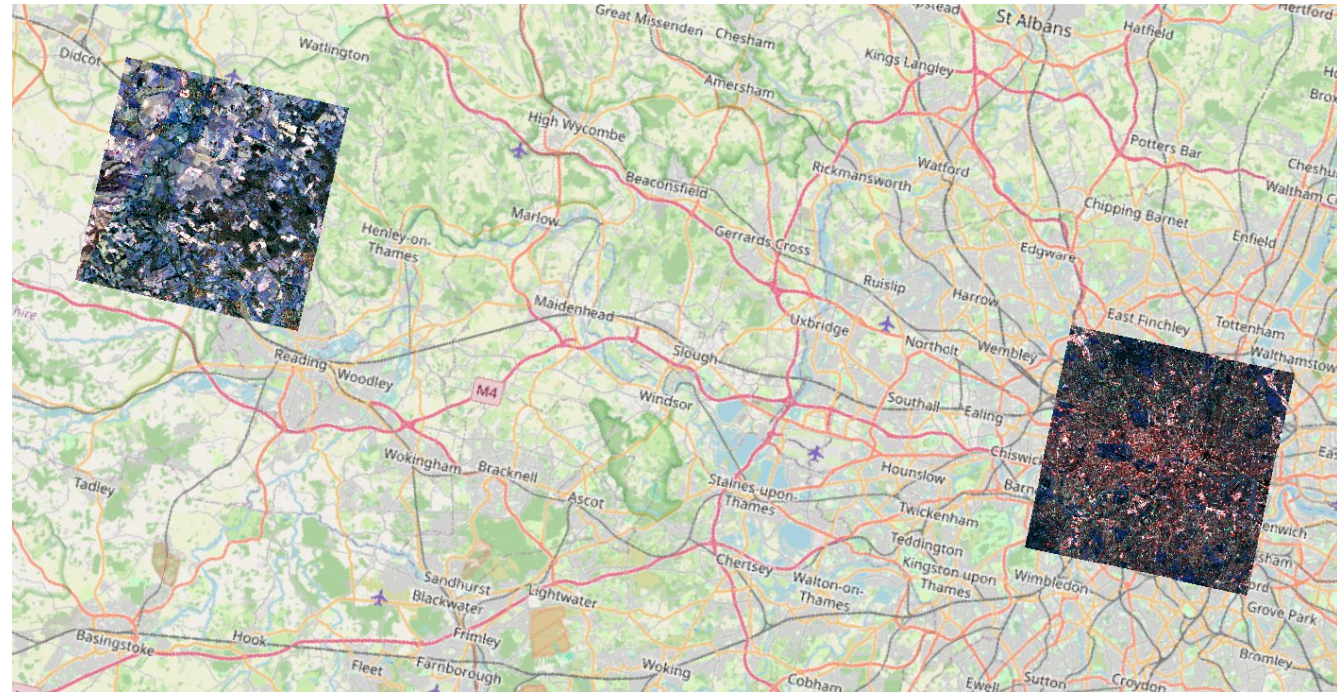
Challenges

Domain Shift

Domain shift occurs when the data used for training and those observed during validation or testing follow **different distributions**.

In satellite imagery, this difference can arise from **geographical variations** and **differences in land-cover characteristics**.

As a result, the **model may struggle to generalize** to geographical areas not seen during training.





Index

- › Context
- › Challenges
- › Approach
- › Proposed Solution
- › Results



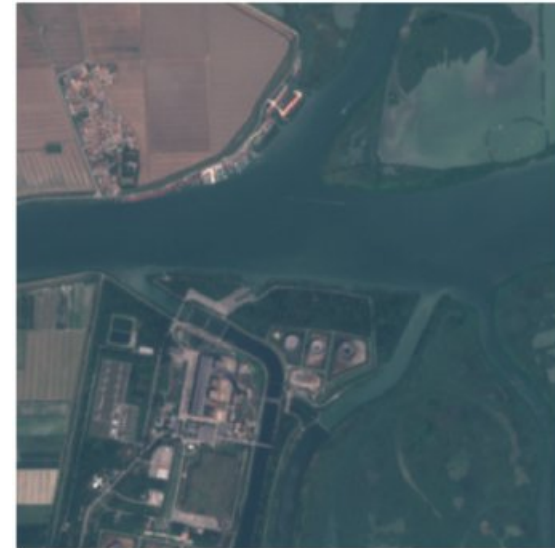
Approach

Dataset SWED

The Sentinel-2 Water Edges Dataset (SWED) is a dataset developed by the UK Hydrographic Office (UKHO) for training and evaluating deep learning models on satellite image segmentation.

Each **satellite image contains 12 spectral bands**, providing information from different regions of the electromagnetic spectrum.

It consists of multispectral Sentinel-2 images paired with pixel-level segmentation masks for binary Water / Non-Water classification.



Multispectral image



Pixel-level Mask

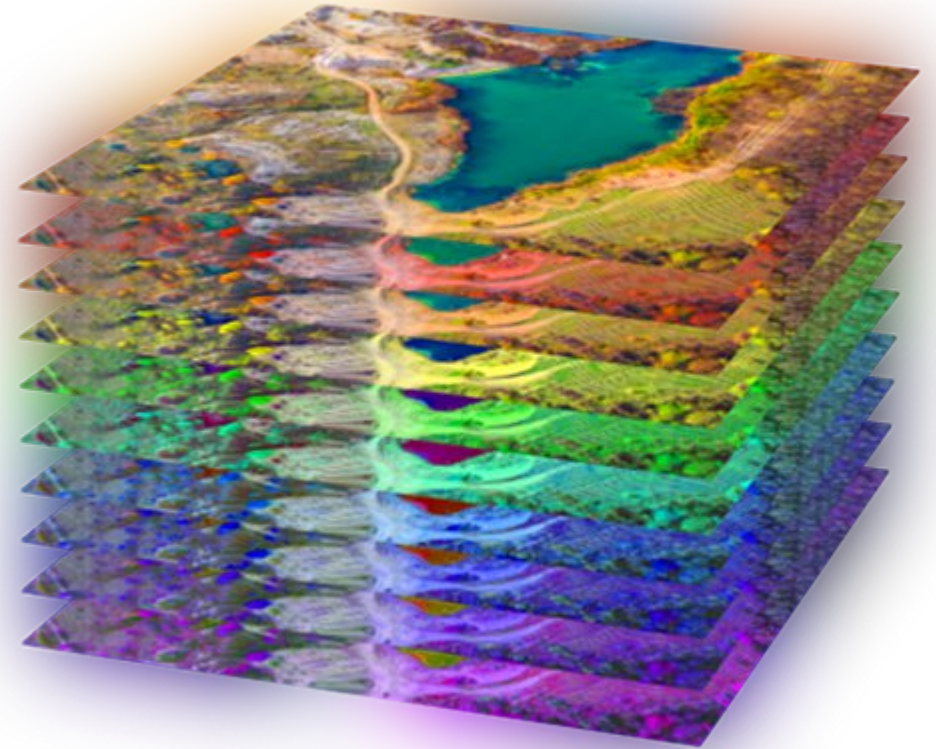


Approach

Multispectral Input

Six spectral bands were selected from the SWED Sentinel-2 imagery to provide the model with complementary information from both the visible and infrared spectrum.

The input combines visible, **Near-Infrared (NIR)**, and **Short-Wave Infrared (SWIR)** information, allowing the network to exploit **spectral differences between land and water**.





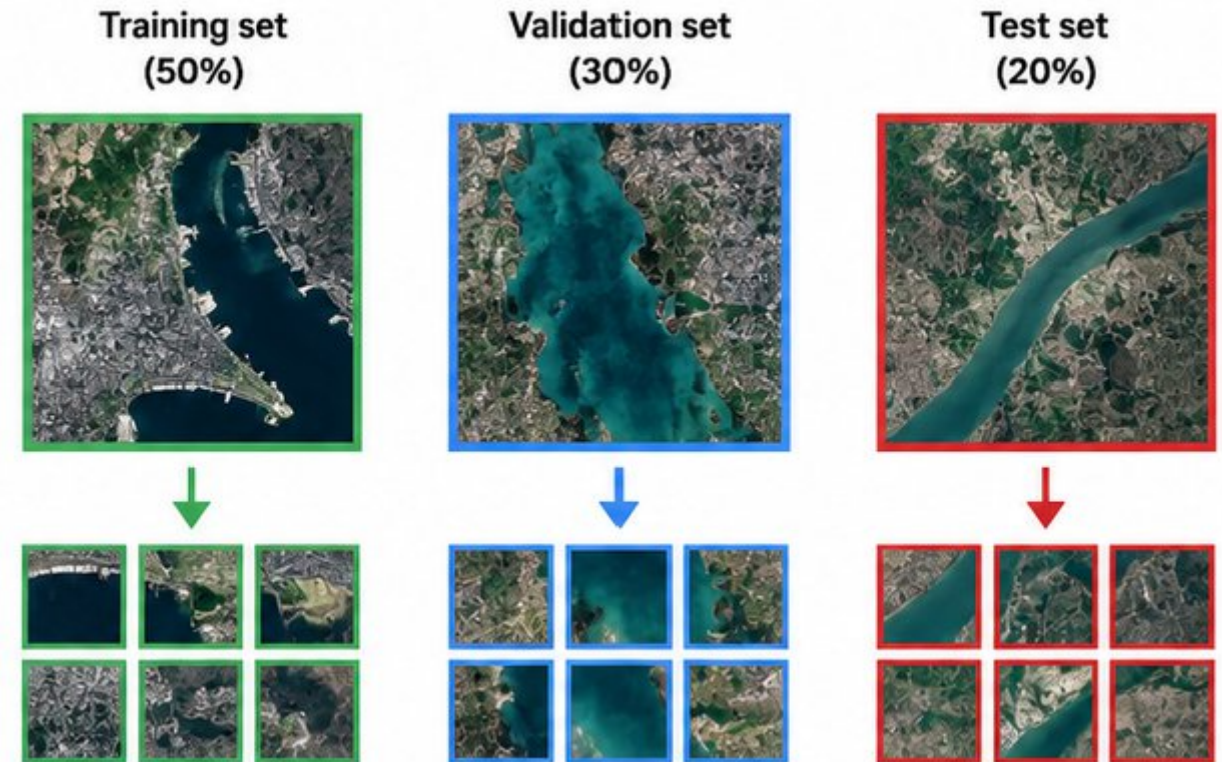
Approach

Dataset Splitting Strategy

To prevent spatial data leakage, the dataset is split at the scene level rather than randomly at patch level.

All 256x256 patches belonging to the same satellite scene are kept within the same subset, ensuring that the same scene never appears across training, validation and test sets.

Scenes are assigned to **Training (50%)**, **Validation (30%)** and **Test (20%)**, while preserving a similar Water / Non-Water pixel distribution across the three subsets.



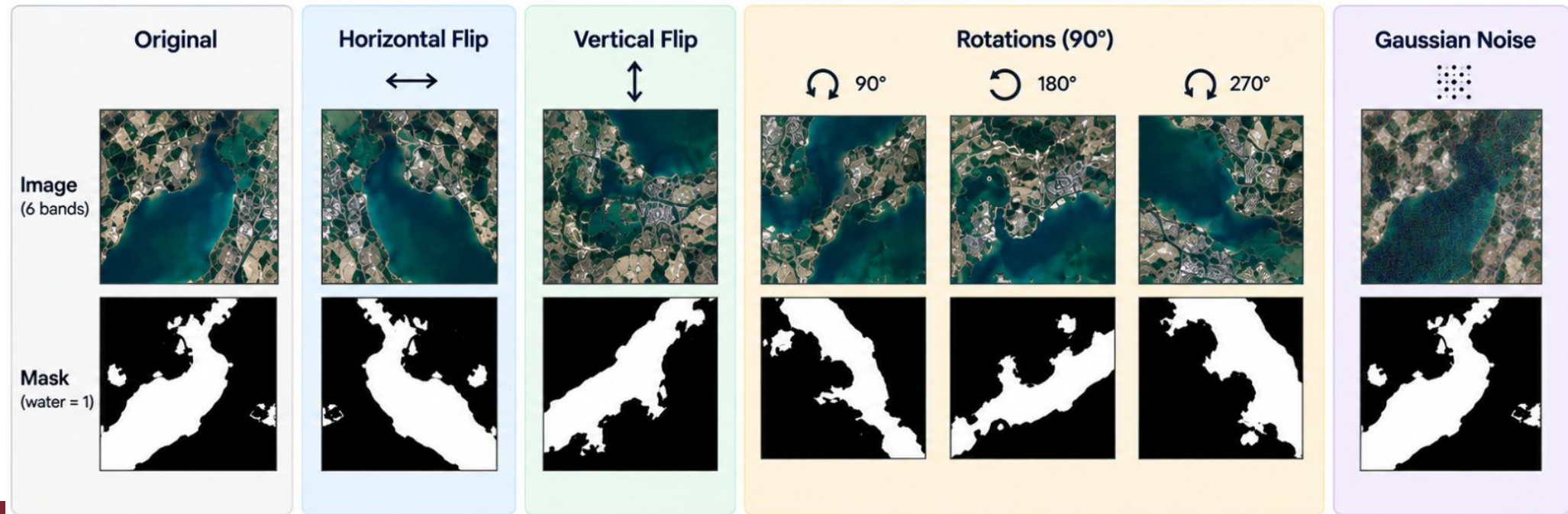


Approach

Data Augmentation

Data augmentation is applied to **increase the variability of the training data and improve model generalization**.

- **Random flips and 90° rotations** introduce different spatial orientations while preserving the semantic content of the scene.
- **Gaussian Noise** introduces small perturbations in the spectral values, making the model more robust to variations in satellite acquisitions.





Index

- › Context
- › Challenges
- › Approach
- › Proposed Solution
- › Results

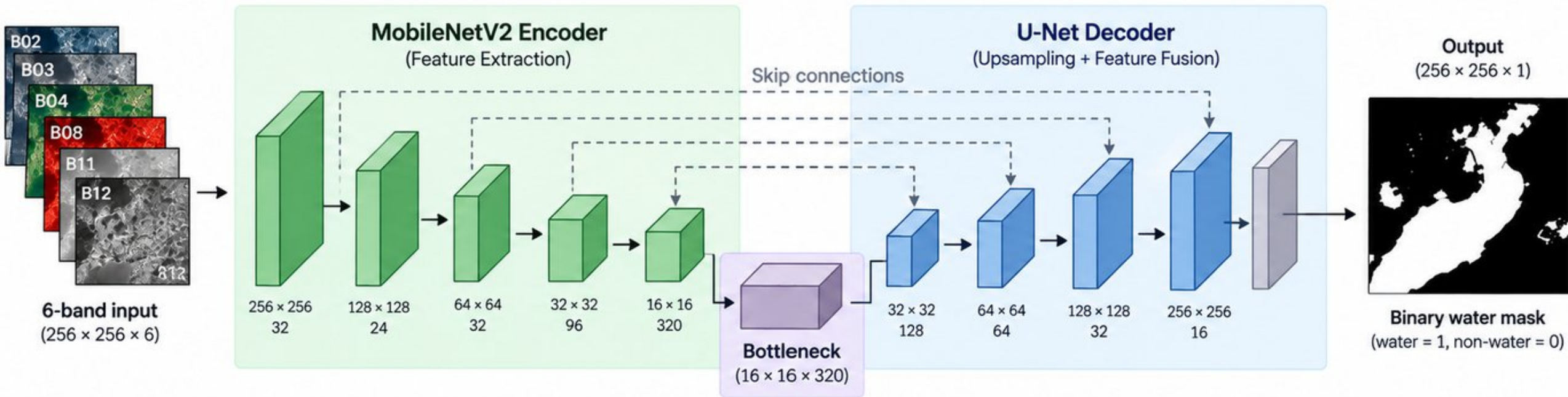


Proposed Solution

U-Net with MobileNetV2 Encoder

The proposed model is based on a **U-Net architecture with a MobileNetV2 encoder**, adapted to process six Sentinel-2 spectral bands.

MobileNetV2 provides a lightweight feature extraction backbone, while the **U-Net architecture enables pixel-level reconstruction** of the Land/Water segmentation mask.





Proposed Solution

How U-Net Works

U-Net follows an encoder-decoder architecture designed to combine semantic understanding with precise spatial localization.

During the **encoding** path, the **image is progressively downsampled to extract increasingly abstract and contextual features**.

During the **decoding** path, these **features are progressively upsampled to recover the original spatial resolution**.

Skip connections transfer high-resolution features from the encoder to the corresponding decoder stages, helping preserve object boundaries and fine spatial details.

This makes U-Net particularly effective for pixel-level segmentation tasks.



Proposed Solution

Training Strategy

The model is trained end-to-end using **Binary Cross-Entropy with Logits** (BCEWithLogits) and the **AdamW optimizer**.

A **ReduceLROnPlateau** scheduler progressively reduces the learning rate when the **validation IoU stops improving**, enabling finer optimization in later training stages.

Early stopping terminates training after 10 epochs without improvement, while the best model checkpoint is selected according to the highest validation IoU.

Training uses a **batch size of 8** with **gradient accumulation over 2 steps**, resulting in an effective batch size of 16.



Index

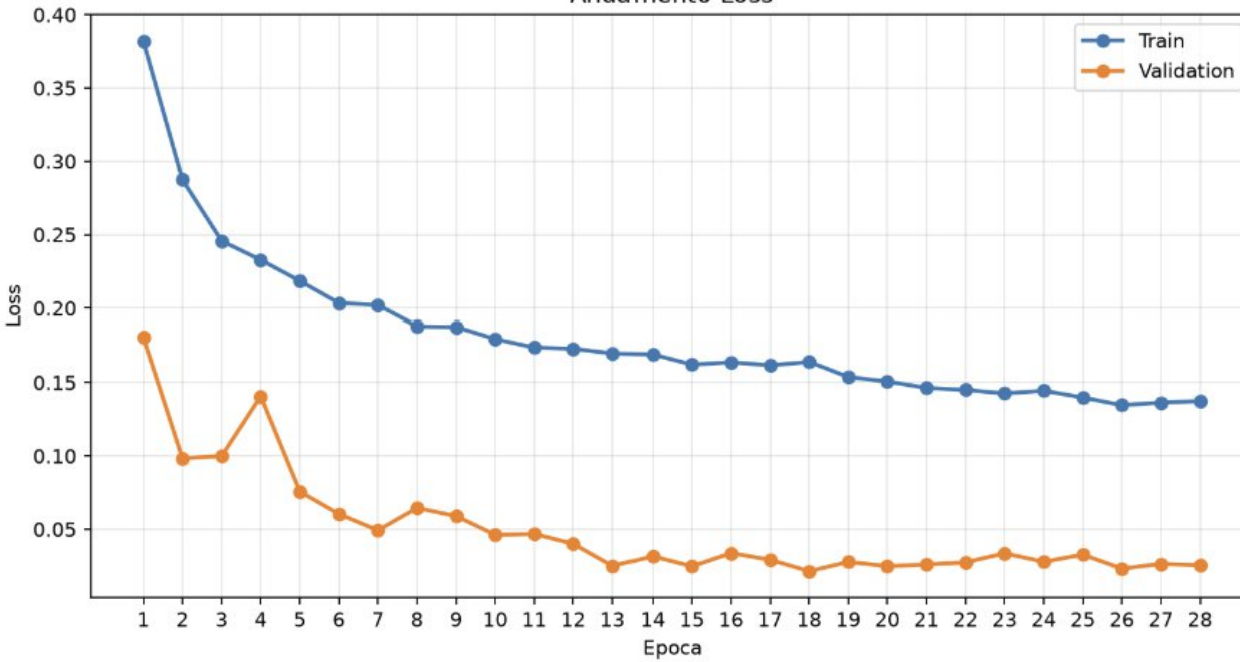
- › Context
- › Issues
- › Approach
- › Proposed Solution
- › Results



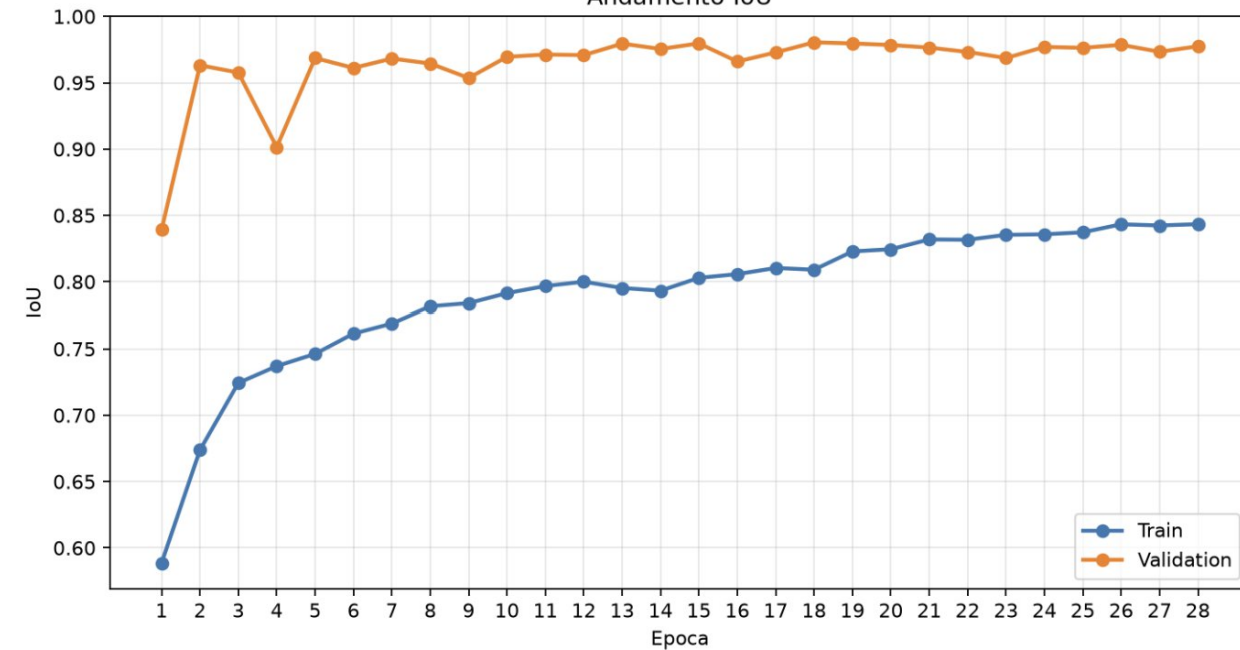
Results

Learning Curves

Andamento Loss



Andamento IoU



Immagini test	IoU	Accuracy	Precision	Recall	Dice score
5231	94%	98%	96%	98%	94%



Results

Semantic Segmentation Tests

Satellite image
RGB: red, green, blue



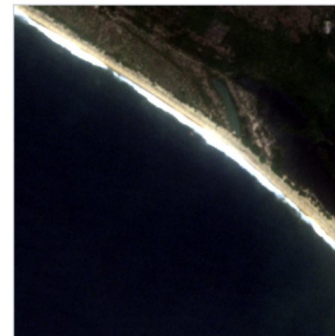
Ground truth
Reference mask



Prediction
Model-predicted mask



Satellite image
RGB: red, green, blue



Ground truth
Reference mask



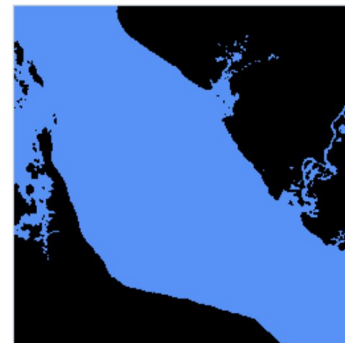
Prediction
Model-predicted mask



Satellite image
RGB: red, green, blue



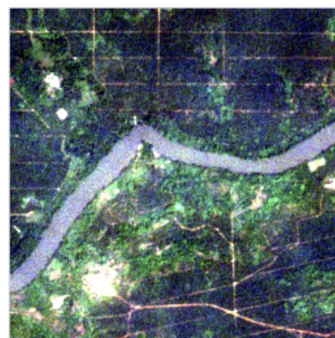
Ground truth
Reference mask



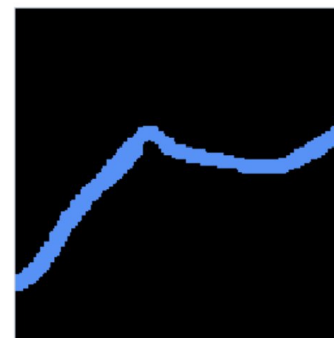
Prediction
Model-predicted mask



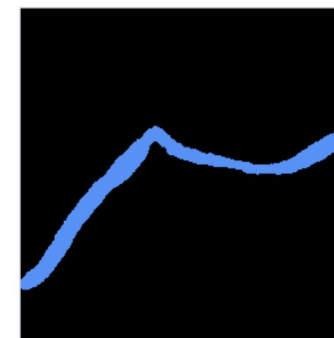
Satellite image
RGB: red, green, blue



Ground truth
Reference mask



Prediction
Model-predicted mask





Results

Semantic Segmentation with Copernicus Images

Out-of-dataset evaluation on unseen Copernicus Sentinel-2 imagery.





Results

Issues with Copernicus Images

When tested on external urban satellite imagery, the model correctly identifies the main water body but produces **several false positives**. This highlights a **domain shift** between the coastal environments represented in SWED and dense urban landscapes unseen during training.

